

Increasing second quotients and asymptotically real zeros

Candidate full proof of the conjecture in arXiv:2008.04754

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Status: candidate full proof, likely valid. If a real entire function has positive Taylor coefficients and $1 < q_2 \leq q_3 \leq \dots$ for its second coefficient quotients, then all but finitely many of its zeros are real and simple. This proves the source's explicit conjecture and removes its lower bound $q_2 \geq 2\sqrt[3]{2}$.

1 Source conjecture

The source is T. H. Nguyen and A. Vishnyakova, *On the entire functions from the Laguerre–Pólya I class having the increasing second quotients of Taylor coefficients*, arXiv:2008.04754, published in *J. Math. Anal. Appl.* 498 (2021), 124959 [1].

For

$$f(z) = \sum_{k=0}^{\infty} a_k z^k, \quad a_k > 0,$$

define

$$q_n = \frac{a_{n-1}^2}{a_{n-2}a_n} \quad (n \geq 2).$$

The source proves that all but finitely many zeros are real and simple when $2\sqrt[3]{2} \leq q_2 \leq q_3 \leq \dots$. It then conjectures the same conclusion under the qualitative assumption $1 < q_2 \leq q_3 \leq \dots$.

2 Proof intuition

Rescale at the n th coefficient peak. Monotonicity of the second quotients makes the normalized Laurent tails uniformly summable, while the rescaled functions converge either to a Jacobi triple-product with simple positive geometric zeros or to $1 - z^{-1}$. Rouché's theorem then produces one simple real zero near every large coefficient scale. A fixed-circle argument-principle count shows that only finitely many zeros are left over, so the produced zeros exhaust the tail of the zero set.

3 Coefficient-peak scaling

Multiplying the function and rescaling its variable by positive constants does not change its second quotients or the reality and simplicity of its zeros. We may therefore assume $a_0 = a_1 = 1$. Put

$$p_n = \frac{a_{n-1}}{a_n} \quad (n \geq 1).$$

Then $p_1 = 1$, $p_n/p_{n-1} = q_n$, and hence

$$p_n = q_2 q_3 \cdots q_n. \tag{1}$$

In our previous work, we have obtained a new necessary condition for an entire function to belong to the Laguerre-Pólya class of type I in terms of the closest to zero roots.

Theorem D (T. H. Nguyen, A. Vishnyakova, [24]). *Let $f(z) = \sum_{k=0}^{\infty} a_k z^k$, $a_k > 0$ for all k , be an entire function. Suppose that the quotients $q_n(f)$ satisfy the following condition: $q_2(f) \leq q_3(f)$. If the function f belongs to the Laguerre-Pólya class, then there exists $z_0 \in [-\frac{a_1}{a_2}, 0]$ such that $f(z_0) \leq 0$.*

Our first result concerns the possible number of nonreal zeros of a real entire function whose sequence of the second quotients of Taylor coefficients is non-decreasing.

Theorem 1.1. *Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$, $a_k > 0$, $k = 0, 1, 2, \dots$, be an entire function such that $2\sqrt[3]{2} \leq q_2(f) \leq q_3(f) \leq q_4(f) \leq \dots$. Then all but a finite number of zeros of f are real and simple.*

In connection with the theorem above, we formulate the following conjecture.

Conjecture. *Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$, $a_k > 0$, $k = 0, 1, 2, \dots$, be an entire function such that $1 < q_2(f) \leq q_3(f) \leq q_4(f) \leq \dots$. Then all but a finite number of zeros of f are real and simple.*

The second result of our paper is the following criterion for belonging to the Laguerre-Pólya class of type I for real entire functions with the regularly non-decreasing sequence of second quotients of Taylor

Figure 1: Theorem 1.1 and the explicit conjecture on page 5 of the source.

Let

$$\varphi(z) = f(-z) = \sum_{k=0}^{\infty} (-1)^k a_k z^k$$

and define, for $z \neq 0$,

$$G_n(z) = \frac{(-1)^n \varphi(p_n z)}{a_n p_n^n z^n}. \quad (2)$$

Lemma 1 (Scaling limits). *The monotone sequence (q_n) has a limit $a \in [q_2, \infty]$. The following local-uniform limits hold on $\mathbb{C} \setminus \{0\}$.*

If $a < \infty$, then

$$G_n(z) \longrightarrow G_a(z) := \sum_{m \in \mathbb{Z}} (-1)^m a^{-m(m+1)/2} z^m. \quad (3)$$

If $a = \infty$, then

$$G_n(z) \longrightarrow G_{\infty}(z) := 1 - z^{-1}. \quad (4)$$

Proof. Expanding (2) around its central coefficient gives

$$G_n(z) = \sum_{m=-n}^{\infty} (-1)^m c_{n,m} z^m,$$

where

$$c_{n,m} = p_n^m \frac{a_{n+m}}{a_n} = \prod_{j=1}^m \frac{p_n}{p_{n+j}} \quad (m \geq 0), \quad (5)$$

$$c_{n,-r} = p_n^{-r} \frac{a_{n-r}}{a_n} = \prod_{j=0}^{r-1} \frac{p_{n-j}}{p_n} \quad (1 \leq r \leq n). \quad (6)$$

Here empty products equal one. Set $c = q_2 > 1$. From $p_k/p_{k-1} = q_k \geq c$ we obtain the uniform bounds

$$0 < c_{n,m} \leq c^{-m(m+1)/2}, \quad 0 < c_{n,-r} \leq c^{-r(r-1)/2}. \quad (7)$$

On every compact annulus $\delta \leq |z| \leq R$, the corresponding two Gaussian-type series are summable. Thus (7) supplies a Weierstrass majorant for both Laurent tails.

If $q_n \rightarrow a < \infty$, formulas (5)–(6) give, for every fixed $m \in \mathbb{Z}$,

$$c_{n,m} \longrightarrow a^{-m(m+1)/2}.$$

Dominated convergence for the Laurent series proves (3). If $q_n \rightarrow \infty$, only $c_{n,0} = c_{n,-1} = 1$ survive; every other fixed coefficient tends to zero. The same majorants prove (4). $\square$

For finite a , put $q = a^{-1} \in (0, 1)$. The Jacobi triple-product identity gives

$$G_a(z) = (qz; q)_\infty (1/z; q)_\infty (q; q)_\infty, \quad (8)$$

where $(x; q)_\infty = \prod_{j=0}^{\infty} (1 - xq^j)$. Consequently the zeros of G_a are precisely

$$\{a^k : k \in \mathbb{Z}\}, \quad (9)$$

and every one is positive and simple. The limit $G_\infty = 1 - z^{-1}$ likewise has the single simple zero 1 on $\mathbb{C} \setminus \{0\}$.

4 Proof of the conjecture

Theorem 1. *Let $f(z) = \sum_{k=0}^{\infty} a_k z^k$ be entire, with $a_k > 0$, and suppose*

$$1 < q_2 \leq q_3 \leq q_4 \leq \cdots.$$

Then all but finitely many zeros of f are real, negative, and simple.

Proof. We use the preceding normalization. First suppose $q_n \rightarrow a < \infty$. Choose $\varepsilon > 0$ so small that the closed disk

$$D = \{z : |z - 1| \leq \varepsilon\}$$

contains no zero of G_a except 1, and also

$$1 + \varepsilon < \sqrt{a}, \quad \frac{1 + \varepsilon}{1 - \varepsilon} < a. \quad (10)$$

Lemma 1 and Rouché's theorem imply that, for every sufficiently large n , G_n has exactly one zero in D , counted with multiplicity. Since G_n is real on the real axis and D is invariant under conjugation, that unique zero is real; it is simple and belongs to $(1 - \varepsilon, 1 + \varepsilon)$. By (2), φ therefore has a simple positive zero

$$x_n \in ((1 - \varepsilon)p_n, (1 + \varepsilon)p_n). \quad (11)$$

The second inequality in (10) and $q_{n+1} \rightarrow a$ show that these intervals are pairwise disjoint once n is large enough.

It remains to show that these zeros account for all but finitely many zeros. Set $r = \sqrt{a}$. By (9), G_a has no zero on $|z| = r$. Local-uniform convergence on that circle implies that, for all large n ,

$$\text{wind}(G_n(re^{it}), 0) = \text{wind}(G_a(re^{it}), 0) =: W,$$

where W is a fixed integer. The meromorphic function G_n has a pole of exact order n at zero, and its other zeros are exactly the rescaled zeros of φ . The argument principle therefore yields

$$N_\varphi(rp_n) = n + W, \quad (12)$$

where zeros in $|z| < rp_n$ are counted with multiplicity.

Choose K large enough that all preceding assertions hold and $K + W - 1 \geq 0$. For every $n \geq K$, the distinct simple positive zeros $x_K, \dots, x_n$ all lie in $|z| < rp_n$, by the first inequality in (10). Comparing their number with (12), at most $K + W - 1$ zeros in that disk are not members of this sequence. Since $p_n \rightarrow \infty$, the disks exhaust the plane. Hence globally only finitely many zeros of φ are not among the simple positive zeros x_n .

If $q_n \rightarrow \infty$, repeat the same argument with the limit $G_\infty = 1 - z^{-1}$ and $r = 2$. Choose $0 < \varepsilon < 1$ with $1 + \varepsilon < 2$. Rouché again gives one simple real zero near every p_n ; the corresponding intervals are eventually disjoint because $q_{n+1} \rightarrow \infty$. On $|z| = 2$, the winding number of $(z - 1)/z$ is zero, so the analogue of (12) is simply $N_\varphi(2p_n) = n$. The same exhaustion argument finishes this case.

Thus all but finitely many zeros of $\varphi(z) = f(-z)$ are positive real and simple. Equivalently, all but finitely many zeros of f are negative real and simple. $\square$

5 Scope and novelty

The theorem proves the explicit conjecture displayed in Figure 1. It strengthens the source's Theorem 1.1 by removing the numerical lower bound on q_2 . It does not settle the separate question of whether extra hypotheses can be removed from the source's criterion ensuring that *every* zero is real.

The local run indexes and bounded web searches on 11 August 2026 used the arXiv identifier, exact title and conjecture, increasing second quotients, later work by the same authors, coefficient-peak scaling, and Jacobi triple-product terminology. They found the source, a 2021 follow-up that bounds the possible number of nonreal zeros, and a 2022 thesis retaining further questions, but no proof of this conjecture and no matching scaling argument. Novelty confidence is moderate because coverage is bounded.

References

- [1] T. H. Nguyen and A. Vishnyakova, *On the entire functions from the Laguerre–Pólya I class having the increasing second quotients of Taylor coefficients*, J. Math. Anal. Appl. 498 (2021), 124959; arXiv:2008.04754. <https://arxiv.org/abs/2008.04754>.